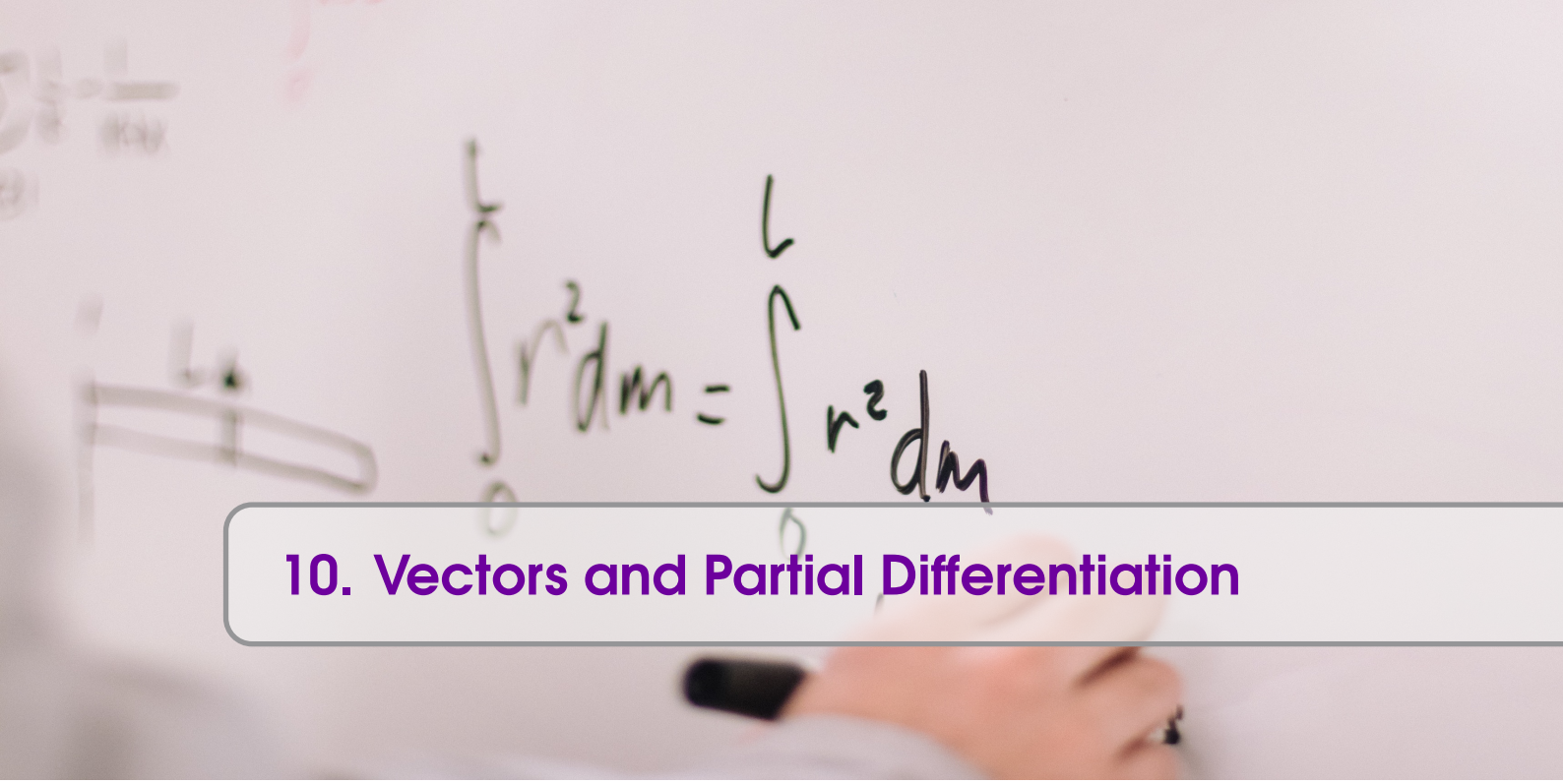


10. Vectors and Partial Differentiation

10.1 Overview

Now that we have concluded the topics Differentiation (Chapters 1–5) and Integration (Chapters 6–9) , we should note that these topics have been centred around a function in one variable or dimension (i.e. $y = f(x)$). We differentiated or integrated in terms of that one variable:

$$\frac{dy}{dx} \text{ or } \int f(x)\,dx$$

In many cases, we may need to work in more than one dimension (e.g. in space) such as $f(x,y) = x^2 + y^2$, therefore, in this Chapter we look at **scalar fields**, **vectors** and **partial differentiation**.

10.2 Scalar fields

In mathematics and physics, a **scalar field** is a function of two or more variables (i.e. $f(x,y)$) that assigns a single scalar value (a number) to every point in a space. Scalar fields represent a single value at different positions. They have magnitude but no direction and can be represented with a colour plot (e.g. temperature over a city, or for example wind speed – but not direction). In essence, they are functions in higher dimensions than we have seen thus far (i.e. $f(x)$). Take a look at the 2-D scalar fields displayed in Figure 10.1.

These figures show three scalar fields: $f(x,y) = x^2$, $f(x,y) = xy$, and $f(x,y) = \frac{1}{2}\left(x^2 + y^2\right)$. Which ones are which? Try to explain why. In the plots white represents high values and black low values. Where are the maximum and minimum values for each scalar field?

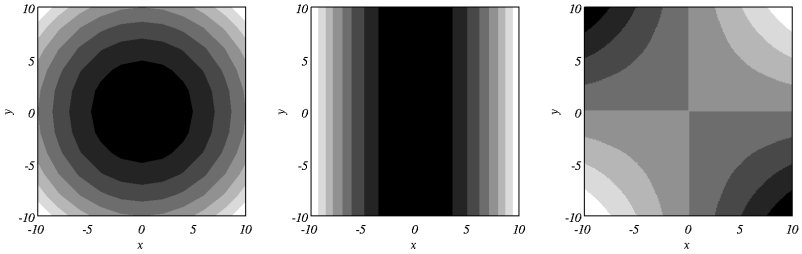


Figure 10.1: Three scalar fields: $f(x,y) = x^2$, $f(x,y) = xy$, and $f(x,y) = \frac{1}{2}\left(x^2 + y^2\right)$

10.3 Vectors

Now take a look at the plots in Figure 10.2 which represent **vector fields**. Can you see the similarity? Imagine these are wind fields, as you might see on a weather forecast. In the vector fields we can determine there is a direction, or flow. The magnitude of that direction can either be taken from the length of the arrow, or by referring to the scalar fields from the plots above. Which vector fields can be aligned to the scalar fields in Figure 10.1?

The left and middle vector field could both relate to the left hand scalar field. So you see here, these vector fields given opposite flow but have the same magnitude as a function of distance. SO how do we represent a vector field algebraically?

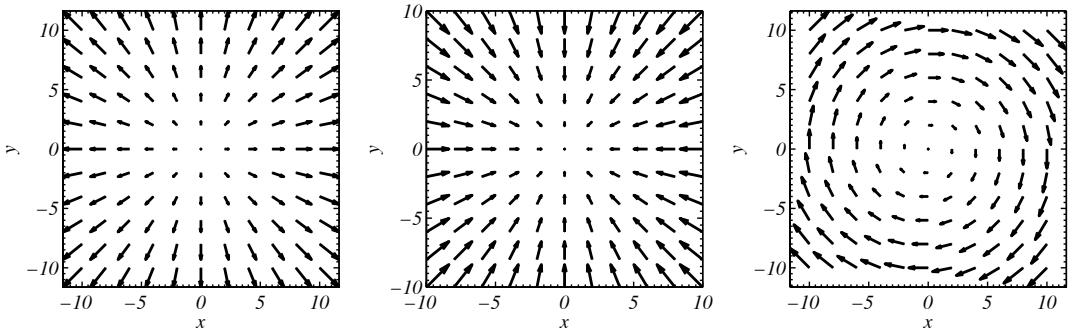


Figure 10.2: Vector fields, from which we can determine there is a direction, or flow. The magnitude of that direction can either be taken from the length of the arrow, or by referring to the scalar fields from the plots in Figure 10.1.

Vector notation

Vectors have **magnitude** and **direction**. We can use a simply graph to illustrate the concept. Consider Figure 10.3. Each line in this figure represents the same vector, but a different starting position. Wherever the starting position is, each vector moves 2 units to the left and 5 units up. If we write this procedure in vector form we write:

$$\vec{v} = \langle -2, 5 \rangle$$

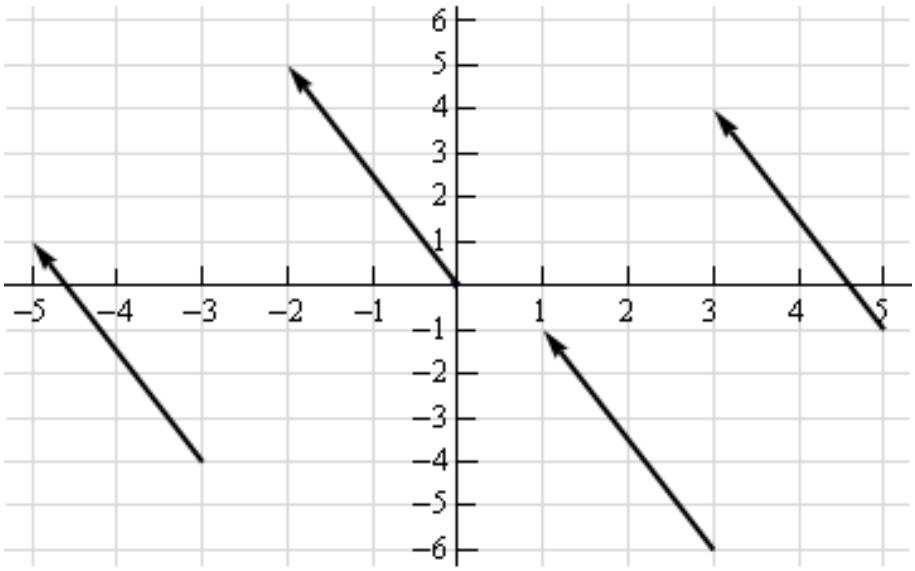


Figure 10.3: Vector fields. In this figure we use the same vector to draw a line but vary the starting position in each case.

Notice the distinction in style between the variable and the brackets. Specifically, the notation $\langle -2,5\rangle$ is fundamentally different from writing $(-2,5)$. The former conveys information about the magnitude and direction of a vector, while the latter represents a point in space defined by coordinates. A vector can be written more generally as:

$$\vec{v} = \langle a_1, a_2\rangle$$

where the constants a_1 and a_2 describe shifts along the horizontal and vertical axes, respectively, based on the assumed coordinate system.

Now, imagine this vector represents movement from point A to B. In this case, we write the vector as $\vec{AB}$. In two-dimensional space, if point A has coordinates $A = (x,y)$ and point B has coordinates $B = (x+a_1, y+a_2)$, the vector $\vec{AB}$ describes the displacement. Note the difference in notation between vectors and coordinates. The same principle extends to three-dimensional systems. If the vector becomes:

$$\vec{v} = \langle a_1, a_2, a_3\rangle$$

then the coordinates of points A and B can be written as $A = (x,y,z)$ and $B = (x+a_1, y+a_2, z+a_3)$, respectively, representing the vector $\vec{AB}$.

More generally, the direction indicated by the arrow over the vector, as in $\vec{AB}$, specifies movement from point A to B along a straight line. If the coordinates of A and B are given as $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$, the vector $\vec{v}$ describing $\vec{AB}$ is:

$$\vec{v} = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3\rangle$$

Conversely, if our vector now starts at B and moves to A, thus $\vec{BA}$, then we can write:

$$\vec{v} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

Using a **start minus end** approach. You can practice these operations in the exercises.

Unit vectors and vector magnitude

In our set of problems we consider vectors in Cartesian form, that is broken down into components along x, y and z of a Cartesian domain. The component along x is component $\hat{i}$, along y is component $\hat{j}$ and along z is component $\hat{k}$. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are all called unit vectors and have length equal to 1.For example, take a look at the following vector:

$$4\hat{i} + 3\hat{j}$$

This tells us that, whatever process is described using this vector, for every movement 4 units along the x axis there is 3 units of movement along the y axis. In our example, $\hat{i}$ and $\hat{j}$ represent what we call a unit vector so has a magnitude of 1. Note that this is equivalent to writing

$$\vec{v} = \langle 4, 3 \rangle$$

from above.

The size of a vector (or its magnitude) can be calculated by taking the square root of all of the square its components (i.e. Pythagoras' theorem). So if $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, then the magnitude, $|\bar{a}|$:

$$|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

A vector that has length that points in the same direction as a general vector $\bar{a}$ is called the unit vector of $\bar{a}$. It is calculated by dividing all components of $\bar{a}$ by the magnitude of $\bar{a}$.

Vector arithmetic

Vectors can be added or subtracted (add or subtract by component) or multiplied by a scalar (constant). For example, Figure 10.4 demonstrates adding two vectors a and b to create a new vector. When adding vectors, you simply add the components for $\hat{i}$ and $\hat{j}$.

Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the addition of the two vectors is given by the following formula:

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

Subtraction is similar. Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the subtraction of the two vectors is given by the following formula:

$$\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

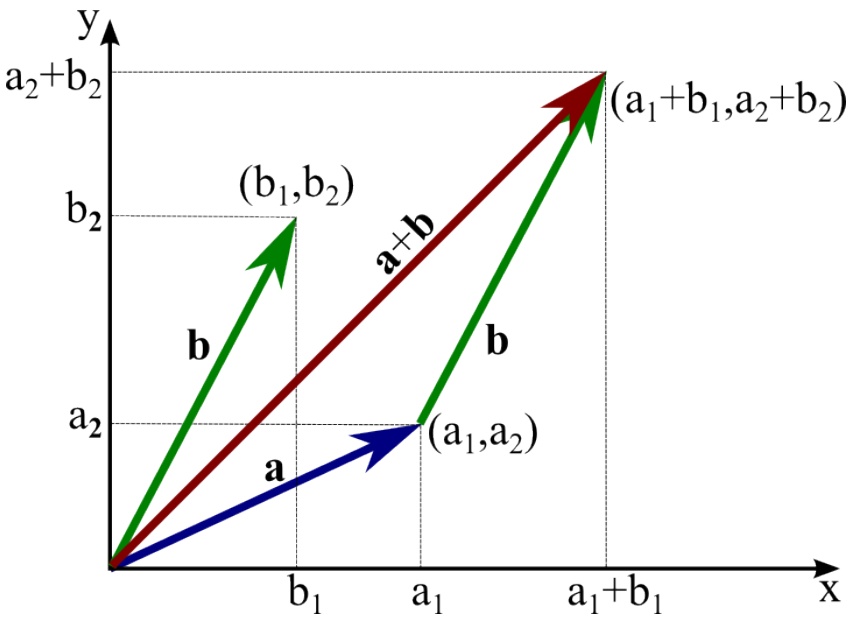


Figure 10.4

This can be harder to visualise. In Figure 10.5 you can see that if we consider we are adding the negative version of vector $\vec{b}$ to $\vec{a}$ (thus $\vec{a}+(-\vec{b})$ since we are performing an extraction), then you can visualise the net direction of travel:

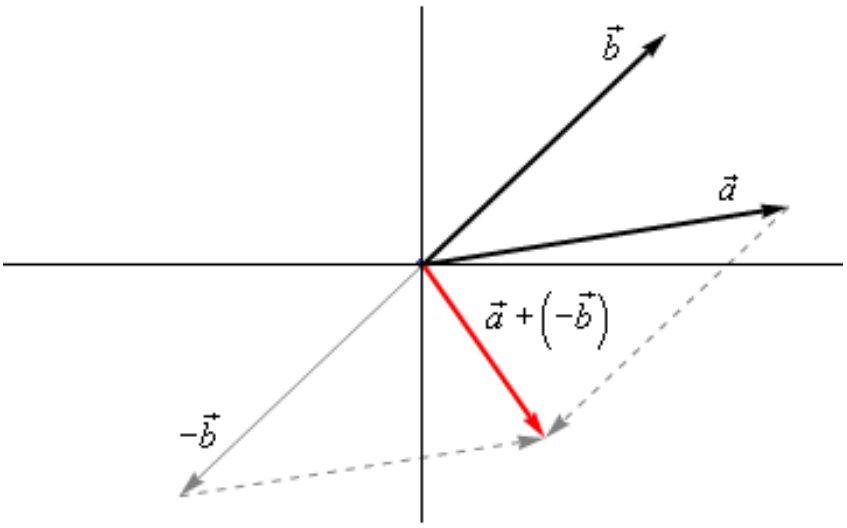


Figure 10.5: Visualisation of vector addition. Credit to https://tutorial.math.lamar.edu/Classes/CalcII/VectorArithmetic.aspx

We can finish on the ability to multiple vectors by a constant. Imagine a situation where a unit vector described the direction of travel every second across a two dimensional grid. If we knew the object was travelling at 15 metes per second, how might we find the absolute position over 1 second? We would multiple the unit vector by the speed. Combined, this new vector represents the velocity of the object. Take a look at Figure 10.6. We have 3

vectors that have been derived from the vector $\vec{a} = \langle 2, 4 \rangle$

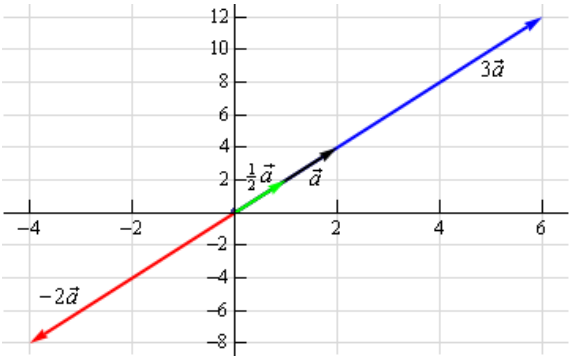


Figure 10.6: Visualisation of vector and scalar multiplication. Credit to https://tutorial.math.lamar.edu/Classes/CalcII/VectorArithmetic.aspx

These lines can be written as the following scalar multiplications:

$$3\vec{a} = \langle 6, 12 \rangle \quad \tfrac{1}{2}\vec{a} = \langle 1, 2 \rangle \quad -2\vec{a} = \langle -4, -8 \rangle$$

Noting that when we multiply our original vector by a negative number, we simply reverse the direction of travel.

10.4 Partial Differentiation

For the vast majority of this course, we have dealt with differentiation of functions in one dimension, or of one variable (i.e. $y = f(x)$). We now think about differentiation of functions of two (or more) variables (i.e. scalar field), $z = f(x, y)$, which now depend on x and y.

For a one dimensional curve the gradient $\frac{dy}{dx}$ is fairly straight forward. When there are more than one dimension we have to use something else to denote the gradient. The standard that is used is called 'grad', which is defined for a scalar field and can be written $grad(f)$ or also written ∇f. We note that ∇f (where $f = f(x, y, z)$ is a scalar field) is actually a vector. In Cartesian coordinates it is:

$$\nabla f \equiv \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} \equiv \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

So the gradient has magnitude and direction. To describe the movement of mass, for example, this is important information. Notice the change in the derivative symbol. Specifically, let us highlight the gradient along the x axis:

$$\frac{\partial f}{\partial x}$$

Up to this point in our course, we have been using the expression:

$$\frac{df}{dx}$$

The change is important. The new expression is called a *partial differential equation*. When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant. We are only interested in information along the axis given by the variable in the denominator. We can get a feel for this through the exercises and solutions provided in the later stages of today's material.

Even when we do not use vector notation, the partial derivatives of a function can be derived. Take the following example:

$$f(x,y) = 2x^2y^3 + y$$

where we have defined a function that varies with respect to both x and y. We have been asked to find the rate of change of the function at a set of coordinates given by (a,b). Conceptually we will need to hold y constant and vary x to determine the gradient along the x axis a vice versa. Indeed, if we were using the method of finite difference, we could use the following assumption:

$$\frac{\partial f}{\partial x}_{(x,y)=(a,b)} = \lim_{h \mapsto 0} \frac{f(a,b+h) - f(a,b)}{h}$$

Of course, we don't need to use finite difference in this instance since we can apply one of our rules of differentiation.

Example 10.1: Let's find the partial derviatives of the following function:

$$f(x,y) = 2x^2y^3 + y$$

If we take y as constant then we can derive the following expression for the partial derivative of f with respect to x:

$$\frac{\partial f}{\partial x} = 4xy^3$$

Note that the second term disappears here as it has no x within it. For the partial derivative of f with respect to y, we arrive at the following:

$$\frac{\partial f}{\partial y} = 6x^2y^2 + 1$$

Putting this into vector form gives us:

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} \\ &= 4xy^3\hat{i} + 6x^2y^2 + 1\hat{j} \end{aligned}$$

or in the form:

$$\begin{aligned}\nabla f &= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \\ &= \left\langle 4xy^3, 6x^2y^2 + 1 \right\rangle\end{aligned}$$

We can use this to calculate the magnitude of the gradient at any given point.

We could go even further and take second partial derivatives of f to obtain get $\nabla^2 f$ which would mean partial differentiating with respect to x and y:

$$\frac{\partial^2 f}{\partial x \partial y}$$

as we are differentiating twice with respect to all combinations of the variables, $\nabla^2 f$ becomes a matrix which has amazing properties in mathematics and physics, but this is outside the remit of this course!

Furthermore, integrating partial differential equations analytically can only be done for simple cases and requires knowledge of Fourier methods, which are outside of the remit of this course. Nearly all modern weather forecast models use approximate methods to find solutions of **partial differential equations**.